

Name: Update Inventory and Notify Shortages

description:

This use case describes the process of updating the inventory after stock changes occur. The inventory worker updates item quantities in the system following actions such as moving items between the warehouse and shelves, adding new items, removing specific items, removing defective items, or removing expired items. The system then checks whether the remaining quantity is below the minimum stock threshold and generates a shortage alert if necessary.

Actors:

Inventory Worker

Pre-Conditions: The relevant ItemType exists in the system catalog.

Post-Conditions:

- The inventory quantities are updated successfully.
- The system verifies the current stock level.
- A shortage alert is generated if the stock level falls below the minimum quantity.
- a purchase request is initiated to the Supplier.
- The updated inventory data is stored in the system.

Basic course of action:

1. The Inventory Worker selects an inventory update operation.
2. The Inventory Worker chooses the relevant item or item type.
3. The Inventory Worker performs the inventory update (e.g., moving items, removing items, adding items, removing defective items).
4. The system updates the inventory quantities.
5. The system checks the current stock level against the minimum stock quantity.
6. The system determines that the stock level is sufficient.
7. The system saves the updated inventory information.
8. The use case ends successfully.

Alternate courses:

a. Stock level is below the minimum quantity

1. The system detects that the stock level is below the defined minimum quantity.
2. The system generates a low-stock alert.
3. The alert is added to the alert list.
4. The system saves the updated inventory information.
5. The system forwards the shortage details to the Supplier to initiate an order.
6. The use case ends successfully.

b. Item or item type does not exist

1. The system cannot find the requested item or item type.
2. The system displays an error message.
3. The inventory update is canceled.

c. Invalid inventory operation

1. The requested operation would result in an invalid inventory state (e.g., negative quantity).
2. The system rejects the operation.
3. The system displays an error message.
4. The use case ends.